Day 3 Reference Guide: Ways to Access AWS, AWS CLI, Global Infrastructure and Shared Responsibility

Cloud Unlock · Day 3 · Services, key ideas and official AWS links

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Live class: Saturday (date TBA), 7:00-9:00 PM IST
Exam focus: Domain 2 (Security and Compliance), Domain 3 (Technology)
Exam tasks: 2.1 · 2.2 · 3.1 · 3.2

Regions, Availability Zones and edge locations were taught on Day 1. Today we revise them in 5 minutes and then use them.

What You Will Know

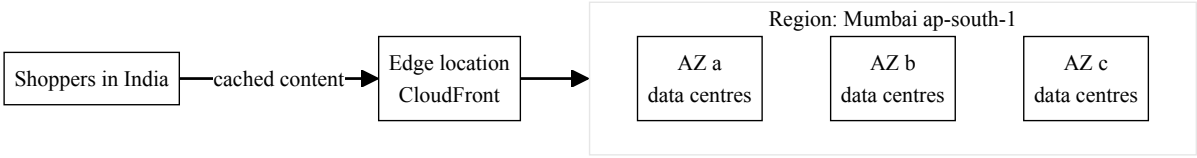
- You can explain Regions, Availability Zones and edge locations, how to choose a Region, and when to use more than one.
- You know the four ways to access AWS, what `aws configure` needs, and ran AWS CLI commands in CloudShell (IAM lists, an S3 bucket).
- You know what AWS secures, what you secure, and what is shared.
- You know where to find AWS compliance reports (AWS Artifact).

Day 3 - Global Infrastructure (Revision), Ways To Access AWS + CloudShell

Exam tasks: 3.1 (ways to access and operate AWS), 3.2 (global infrastructure)

Terms (revision: taught on Day 1)

- **Region**: geographic area with multiple isolated locations (Mumbai `ap-south-1`).
- **Availability Zone (AZ)**: one or more data centres in a Region with separate power and networking.
- **Edge location**: site close to users that caches content (CloudFront).
- **Local Zone / Wavelength Zone**: compute closer to a city / inside 5G networks.



Availability Zones are physically separate, with their own power, cooling and networking, connected by fast private links. They **do not share a single point of failure**, so running in two or more AZs keeps you up if one AZ fails.

Live counts are on aws.amazon.com/about-aws/global-infrastructure. Do not memorise numbers; check the live page.

How To Choose A Region

1. Compliance and data residency
2. Latency to users
3. Service availability
4. Cost

When To Use More Than One Region

- Disaster recovery and business continuity (a whole Region has a problem)
- Low latency for users on different continents
- Data sovereignty (each country's data stays in that country)

Exam pointer: data must stay in a country = choose a Region there. Users far away = CloudFront. High availability = multiple AZs. Region-wide disaster recovery = multiple Regions.

Four Ways To Access AWS

Way	What it is	Example
AWS Management Console	Web browser, click to use	What you used on Day 1
AWS CLI	Commands in a terminal	`aws ec2 describe-regions` in CloudShell
AWS SDKs	Call AWS from your code (Python, Java, JavaScript...)	A Python app uploading to S3 with `boto3`
Infrastructure as Code	Describe resources in a file, AWS builds them	AWS CloudFormation, AWS CDK

All four call the same AWS APIs underneath.

Exam pointer: repeatable, automated deployments = Infrastructure as Code (CloudFormation). Access from application code = SDK.

One-time or repeatable?

- One-time task, such as exploring or checking a setting: the Console is fine.
- Same thing many times, or in many accounts or Regions: use the CLI, scripts or IaC.

The AWS CLI And Access Keys

- On your own laptop, `aws configure` asks for an **access key ID**, **secret access key**, **default Region** and **output format**.
- Access keys are for the CLI and code only. Never share them, never put them on GitHub, never create them for root.
- **AWS CloudShell** is a terminal in the browser with the CLI already installed. It uses the user you are signed in with, so **no access keys** are needed.
- `aws sts get-caller-identity` shows who you are signed in as. Its output contains your account ID, so never share it unmasked.

Official Links

Topic	Link
AWS global infrastructure	https://aws.amazon.com/about-aws/global-infrastructure/
Regions and Availability Zones	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-regions-availability-zones.html
Services available in each Region	https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/
Availability Zone IDs	https://docs.aws.amazon.com/ram/latest/userguide/working-with-az-ids.html
Amazon CloudFront (edge locations)	https://aws.amazon.com/cloudfront/

Topic	Link
AWS CloudShell	https://docs.aws.amazon.com/cloudshell/latest/userguide/welcome.html
AWS CLI	https://aws.amazon.com/cli/
AWS CLI user guide (aws configure)	https://docs.aws.amazon.com/cli/latest/userguide/cli-chap-welcome.html
AWS SDKs and tools	https://aws.amazon.com/developer/tools/
AWS CloudFormation (Infrastructure as Code)	https://aws.amazon.com/cloudformation/
AWS Health	https://docs.aws.amazon.com/health/latest/ug/what-is-aws-health.html

Day 3 - Shared Responsibility Model

Exam tasks: 2.1 (shared responsibility), 2.2 (compliance and governance)

> *AWS secures the cloud. You secure what you build in the cloud.*

	AWS	**You**
Headline	Security **of** the cloud	Security **in** the cloud
Examples	Data centres, hardware, hypervisor, global network	Data, IAM, firewall rules, EC2 guest OS patching

Responsibility Shifts By Service

Service	**AWS handles**	**You handle**
EC2	Hardware, virtualisation	Guest OS, patches, apps, security groups, data
RDS	Hardware, OS, database patching	Data, access, network access, settings
Lambda	Servers, OS, runtime	Code, permissions, data
S3	Hardware, service durability	Bucket permissions, encryption choice, content

Controls: Inherited, Shared, Customer

Type	**Meaning**	**Examples**
Inherited	AWS does it fully; you inherit it	Physical and environmental security of data centres
Shared	Both do their part, each in its own layer	**Patch management** (AWS patches infrastructure, you patch your guest OS and apps), **configuration management**, **awareness and training**
Customer-specific	Only you	Your data, who has access, which Region stores it

Exam pointer: the more managed the service (EC2 -> RDS -> Lambda), the less you manage.

Compliance: Where To Find The Proof

- **AWS Artifact**: free, self-service portal for AWS compliance reports (such as ISO and SOC) and agreements.
- **AWS Compliance Programs** (aws.amazon.com/compliance): which standards AWS meets by industry and country.
- AWS being compliant does **not** make your app compliant. You still configure your part.

Official Links

Topic	**Link**
Shared responsibility model	https://aws.amazon.com/compliance/shared-responsibility-model/

Topic	Link
AWS Artifact	https://docs.aws.amazon.com/artifact/latest/ug/what-is-aws-artifact.html
AWS compliance programs	https://aws.amazon.com/compliance/programs/
AWS Cloud security	https://aws.amazon.com/security/
